AAYUSH DAYAL

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EDUCATION

University of British Columbia — September 2021 – *April 2026*
Bachelor of Applied Science – Mechanical Engineering

PROFESSIONAL EXPERIENCE

Telix – ARTMS INC. — **Kelowna, BC**
MECHANICAL ENGINEERING CAPSTONE STUDENT — ***September 2025 – April 2026***
Target Feedback & Detection System for ARTMS QUANTM Irradiation System (QIS)

- Designed modular polycarbonate hose coupler to integrate multi-sensor PCB within conduit's spatial constraints.
- Engineered pneumatic test rig operating at 110 PSI to validate target detection under simulated operational conds.
- Achieved ISO 7 cleanroom standards using sealed coupling with Buna-N O-rings to prevent air leakage.
- Assist dev. of checkpoint sensors with CAN-bus protocol to relay data & real-time HMI integration on Siemens PLC.
- Delivered $3000 budget-compliant product, prototyping design iterations, leading client briefings,& preparing BOMs.

Vice-Principal Research & Innovation Office - UBC — **Kelowna, BC**
PROJECT MANAGER — ***May 2025 – August 2025***

- Liaised with cross-functional stakeholders across research teams to align project deliverables & requirements.
- Conduct space audits & taxonomical equipment classification for 1000+ capital assets using Excel & MS Access.
- Engineered Access database with ERD design, SQL queries, VBA programming, & automated validation controls.
- Create technical documents, status reports, risk assessments supporting multi-disciplinary research facility planning

Ranoson Springs PVT. LTD — **New Delhi, India**
CAD INTERN — ***May 2023 – August 2023***

- Collaborated on 3D models/assemblies on SolidWorks & maintained ASME Y14.5 standards for timely completion.
- Optimized tooling design & automated conveyor system using ISO 9001 procedures, expediting production by 15%.
- Performed quality control validation using GD&T, ensuring compliance with client contractual & spec. requirements.
- Updated 25+ legacy blueprints with revised specification, eliminating 90% of technical CAD drawing discrepancies.

TECHNICAL PROJECTS

Two-Stage Helical Gearbox Design & Analysis for Industrial Power Transmission — ***April 2024***

- Designed compound reverted gearbox achieving 12 HP transmission with 9:1 reduction ratio & bearing selection.
- Performed stress analysis using Goodman criteria / FEA achieving safety factors > 2 for shafts, gears, & bearings.
- Led iterative design reviews, prototype development, & assist in technical documentation; CAD model & part specs.

Advanced Multi-Component Manufacturing & Assembly: Precision-Engineered Scale Train Model — ***December 2023***

- Developed comprehensive CAD & assembly sequences for 15 interconnected components for 0 assembly conflicts.
- Operated 3-axis CNC mills, engine lathes, waterjet cutting, machining tools & conducted QC using RCCA protocols.
- Tested precision assembly of mechanical interfaces, validating fit & function requirements through CMM inspection.

LEADERSHIP EXPERIENCE

OK MOTORRAD MotoStudent Design Team — **Kelowna, BC**
AERODYNAMICS TEAM LEAD — ***August 2025 – April 2026***

- Designed bike fairing, drafted CAD model, performed FEA & aeroelastic deflection analysis using SolidWorks Suite.
- Led CFD analysis using ANSYS Fluent achieving 15% drag reduction through virtual wind tunnel testing.

The Okanagan Engineering Society — **Kelowna, BC**
VICE PRESIDENT - INTERNAL (VPI) — ***July 2023 – May 2024***

- Liased b/w engineering clubs & the School of Engineering, managing logistics, budgeting, & sponsor engagement.
- Led delegation of engineering students as Head Delegate at multiple engineering conferences across Canada.

TECHNICAL SKILLS

Mechanical	SolidWorks, Rapid Prototyping, FEA, MATLAB, ANSYS Fluent CFD
Manufacturing	ANSYS Granta, GD&T, DFM/DFA, ISO / ASTM Standards, Six Sigma Methods
Other	Engineer-in-Training (EIT) Eligibility, Project Management, MS Office Suite

AWARDS

Okanagan Engineering Competition 2025 – 1st Place, Re-Engineering Category — **Kelowna, BC**